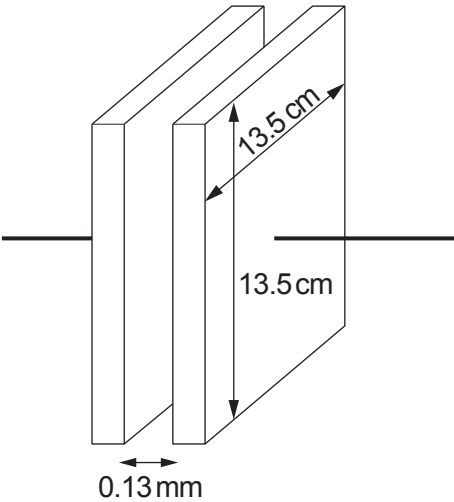


Examiner
only

4. (a) Calculate the capacitance of the capacitor shown.

[2]



.....

.....

.....

.....

.....

.....

(b) (i) Calculate the pd across the terminals of a 5.0 mF capacitor when it stores 1.00 J of energy.

[2]

.....

.....

.....

.....

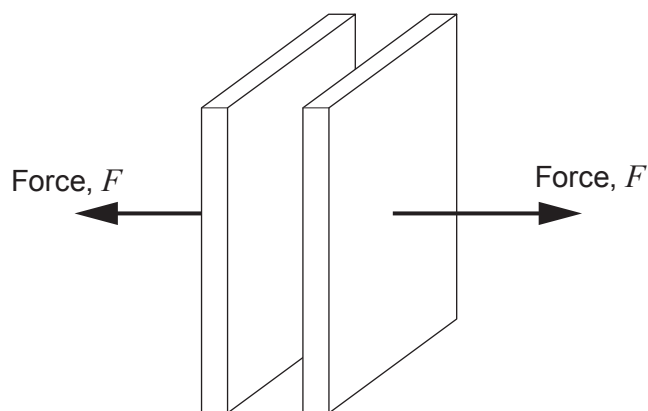
.....

.....



- (ii) In the space provided, draw a diagram to show how you would combine three 5.0 mF capacitors to produce a capacitance of 7.5 mF. [2]

- (c) The separation of the plates of a charged capacitor is increased by the application of a force, F . The capacitor is **isolated** so the charges on the plates remain unchanged.



- (i) State why a force must be exerted to separate the charged plates. [1]

.....

.....

- (ii) Explain why the capacitor stores more **energy** when the separation of the plates is increased even though the charge remains constant. [2]

.....

.....

.....



Examiner
only

(iii) Show that the energy stored by the capacitor is given by:

$$\text{energy stored} = \frac{1}{2} \frac{Q^2 d}{\epsilon_0 A}$$

where Q is the charge stored, d is the separation of the plates, ϵ_0 is the permittivity of free space and A is the area of the plates. [2]

.....

.....

.....

(iv) Bethan states that the force, F , required to separate the plates is given by:

$$F = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$$

Determine whether Bethan is correct to arrive at this conclusion. [2]

.....

.....

.....

13

